

SUPPLEMENTARY INFORMATION FOR

Disentangling the drivers of parasite occurrence in a native and a non-native small mammal across a land-use gradient

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Table S1: Summary of variables used in the final Hierarchical Modeling of Species Communities (HMSC) model. See **SI note 3** for detailed explanations on sampling and measurement methods.

Category	Variable Name	Effect Type	Variable Class	Scale	Explanation
Host species	Host species	Fixed	Factor	Individual	<i>R. rattus</i> / <i>M. brevicaudata</i>
Host attributes	Body size	Fixed	Continuous	Individual	Normalized structural size (hindfoot length)
	Body condition	Fixed	Continuous	Individual	Body mass relative to structural size (residual index)
	Sex	Fixed	Factor	Individual	Male / Female
Microbiome	Diversity	Fixed	Continuous	Individual	Shannon diversity index
	Uniqueness	Fixed	Continuous	Individual	Weighted UniFrac distance to conspecifics
	FB ratio	Fixed	Continuous	Individual	Firmicutes/Bacteroidota ratio
Environment	Land-use	Fixed	Continuous	Individual	Distance from capture location to nearest village center (land-use proxy)
	Season	Fixed	Factor	Season	Sampling period (before wet, after wet, and during dry seasons)
	Random: site	Random	Factor	Site	Unmeasured site-specific factors
Interaction	Host species × Land-use	Fixed	—	Individual	Species-specific response to the land-use gradient
Individual random effect	Random: individual	Random	Factor	Individual	Unmeasured individual-specific factors

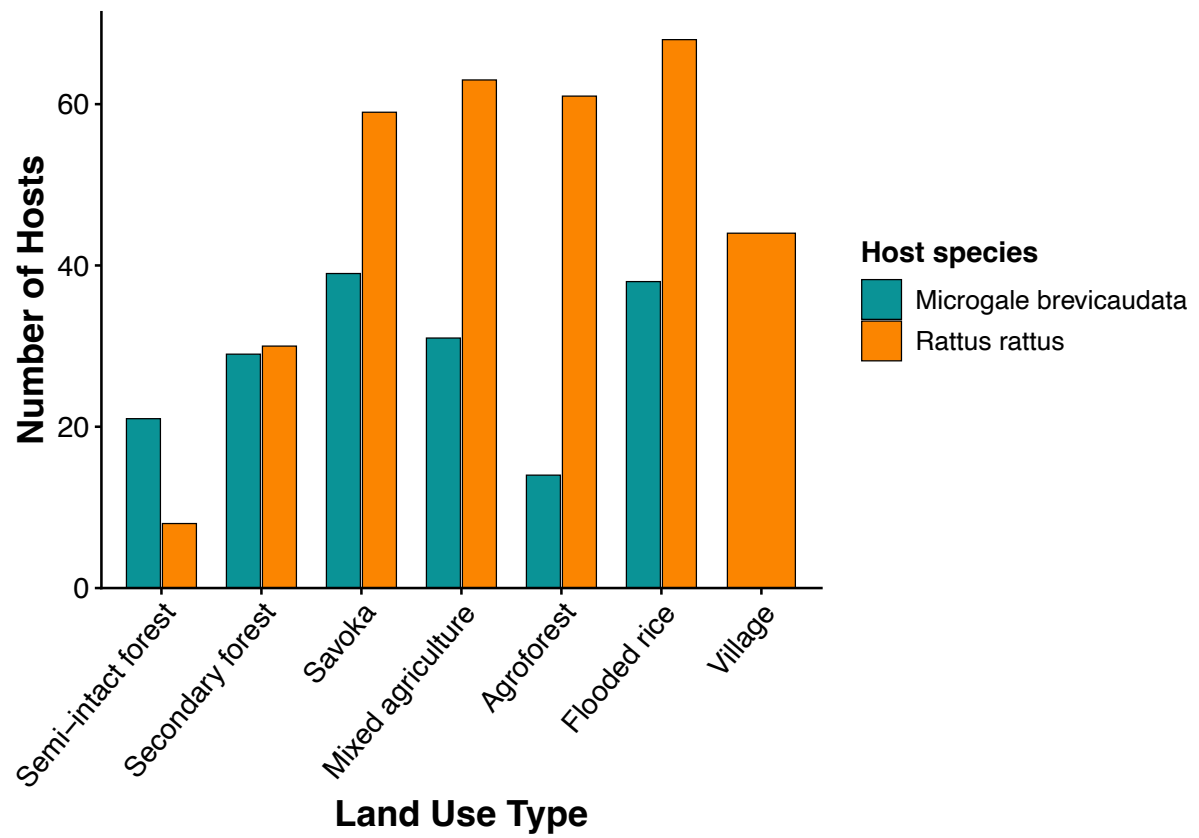


Figure S1: Abundances of *Rattus rattus* and *Microgale brevicaudata* hosts across sites. The graph presents the total number of individuals captured in each site.

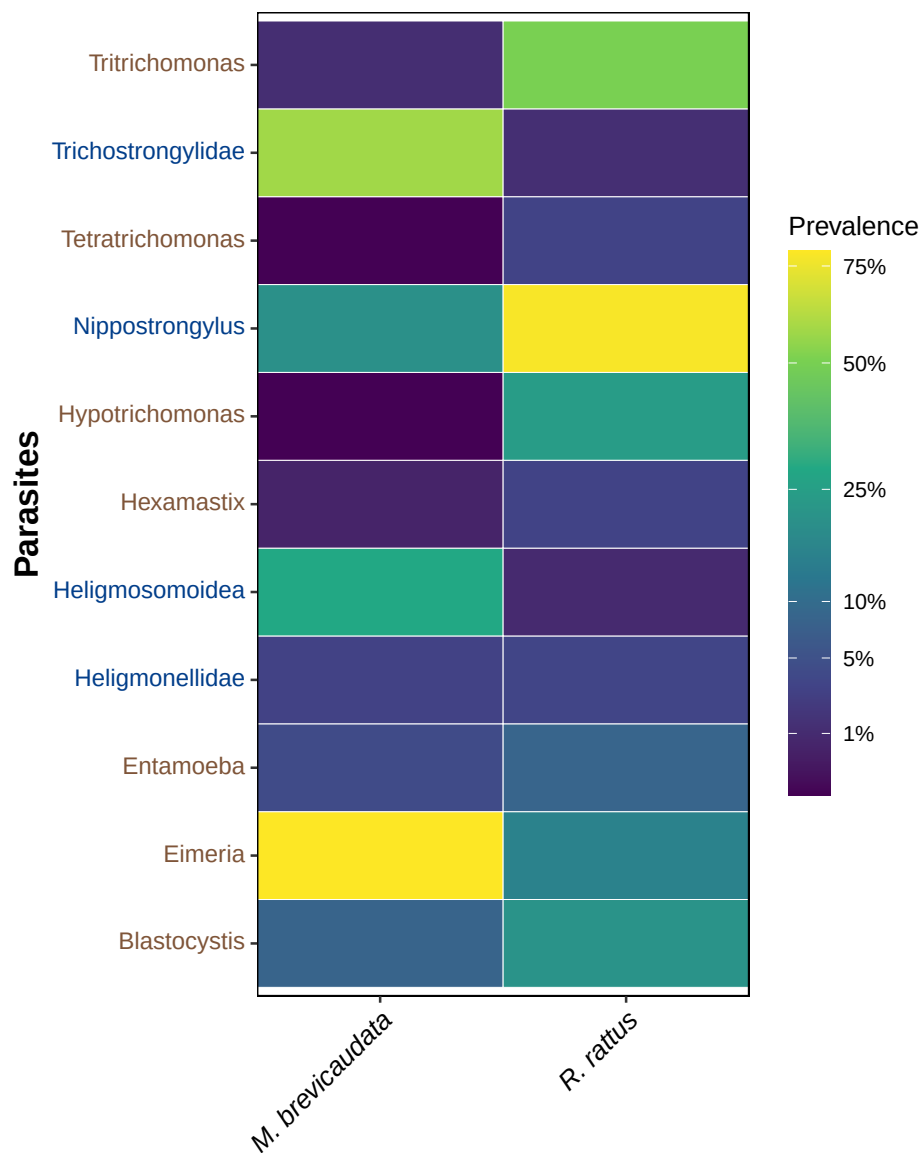


Figure S2: Prevalence of parasite taxa in *Rattus rattus* and *Microgale brevicaudata* hosts. The heat map displays the prevalence of parasites, with the color intensity representing the proportion of infected hosts. Parasite taxa on the y-axis are color-coded by taxonomic group: blue indicates nematodes and brown indicates protozoans.

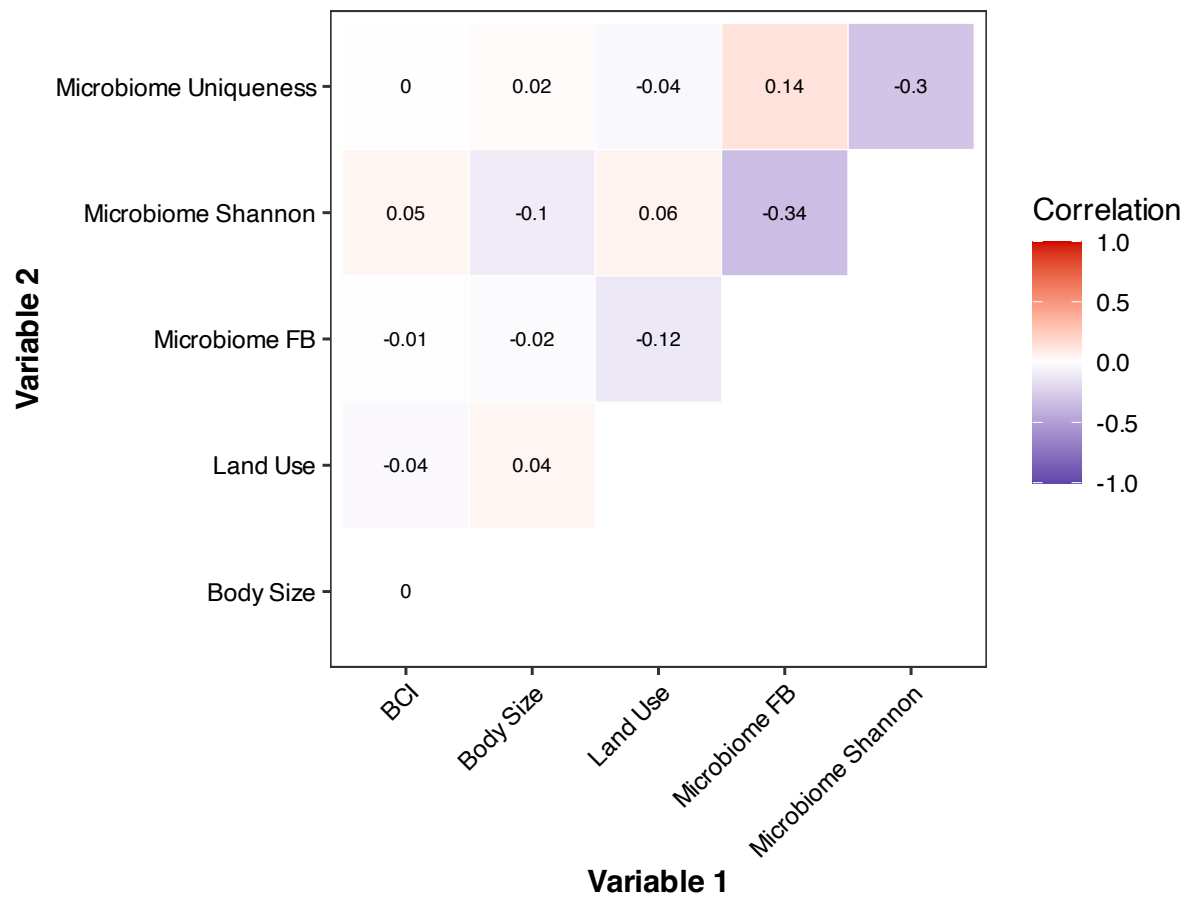


Figure S3: Correlations between predictors. Pairwise Pearson correlations were computed between all continuous predictors. Colors indicate the strength and direction of the correlations: red indicates positive relationships, and purple indicates negative relationships.

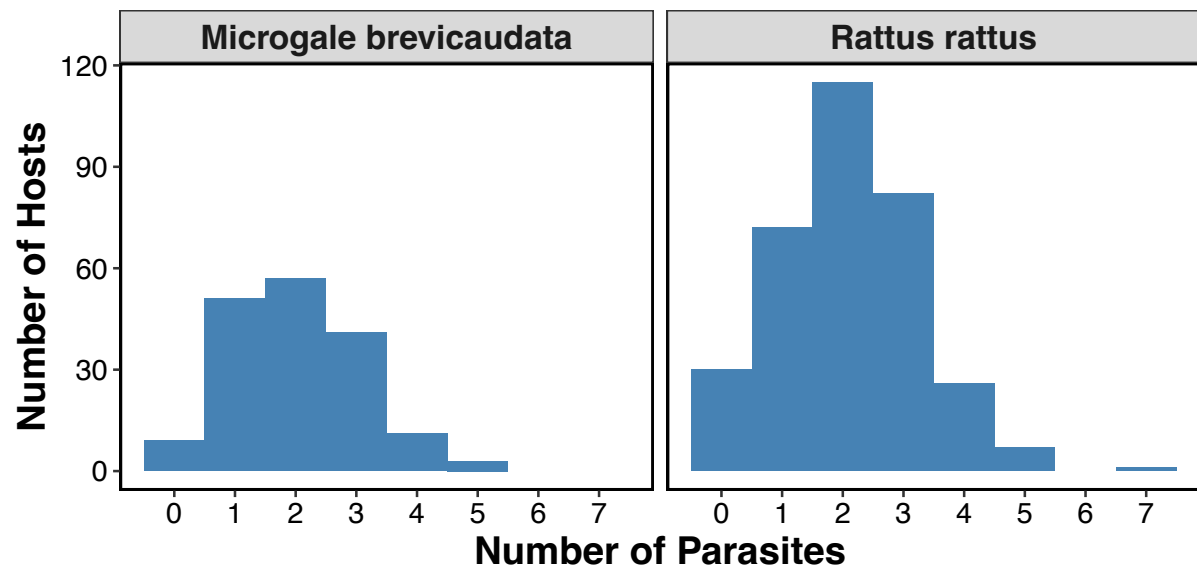


Figure S4: Frequency distribution of parasite species richness per host. Histograms represent the number of host individuals harboring a specific count of parasite taxa.

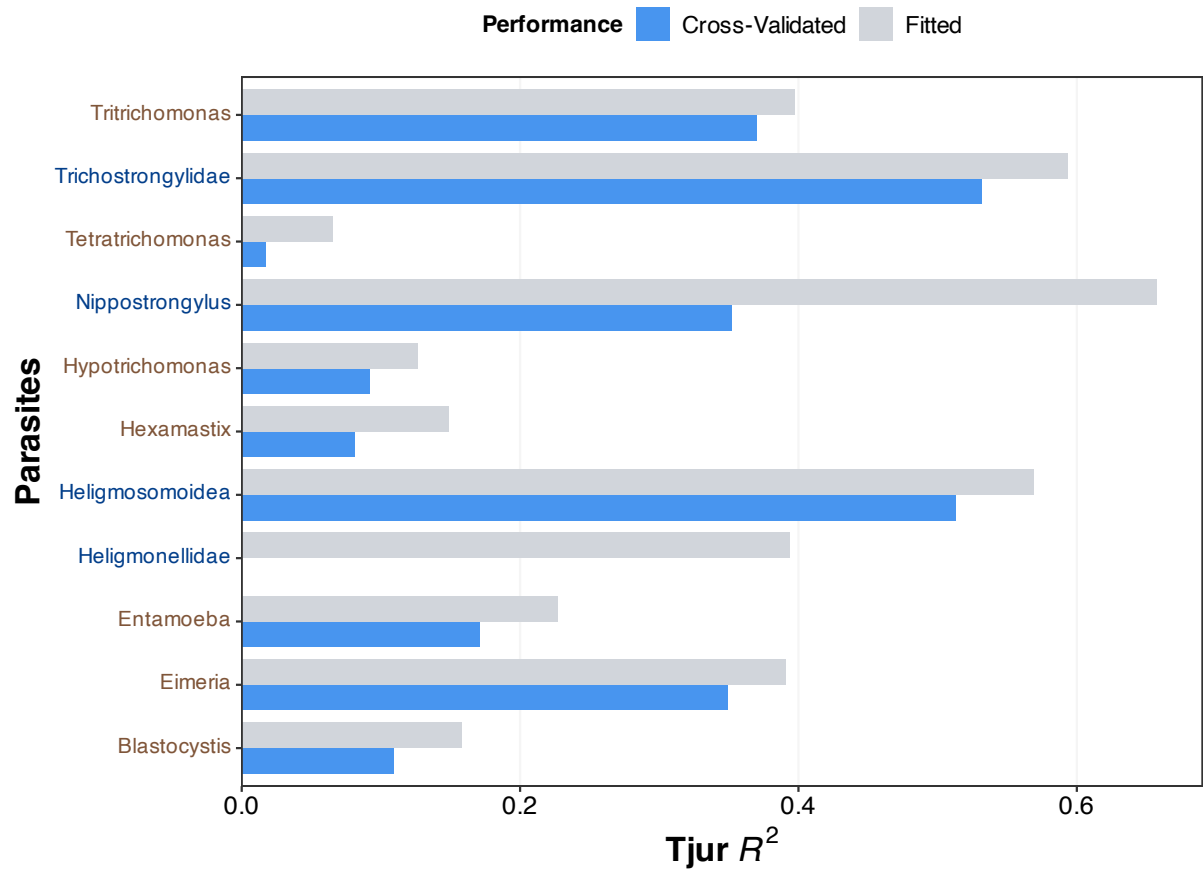


Figure S5: Model performance across parasites. The bar plot illustrates the explained variation ($T_{jur} R^2$) for each parasite, distinguishing between the *Fitted* (in-sample) performance and the *Cross-Validated* (out-of-sample) performance. Parasite taxa on the y-axis are color-coded by taxonomic group: blue indicates nematodes and brown indicates protozoans.

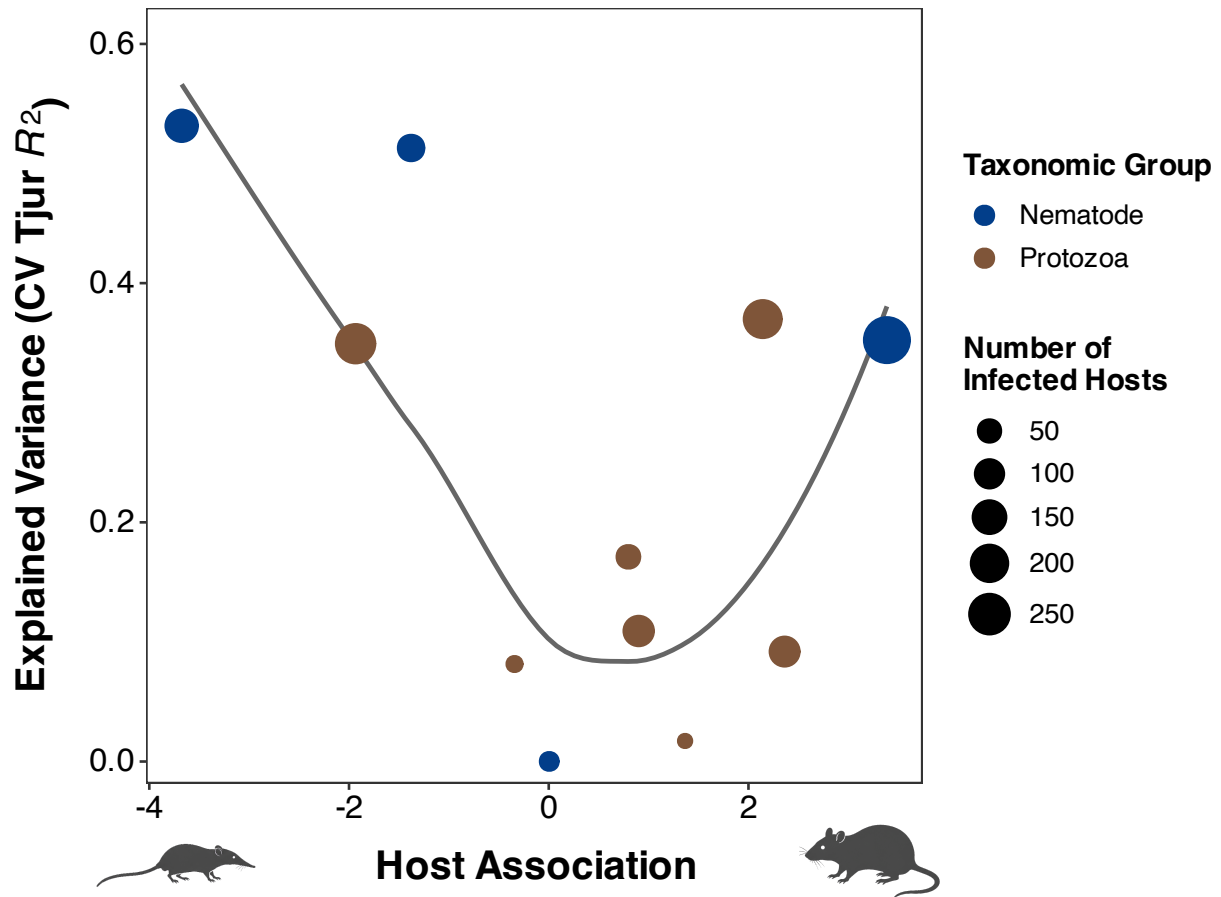


Figure S6: Model predictive performance across parasites. Cross-validated Tjur's R^2 (predictive performance of the HMSC model) as a function of host association. Positive host-association values indicate a stronger association with *Rattus rattus*, negative values a stronger association with *Microgale brevicaudata*, and values near zero indicate no strong host association. Points represent parasite taxa (protozoans in brown, nematodes in blue), sized by the total number of hosts they infect. The trendline is an LOESS smooth fitted for visualization only and does not represent predictions from a statistical model. Given the small number of taxa ($n = 11$), relationships are descriptive. Predictive performance increased with the number of infected hosts (Spearman $\rho = 0.68$, $p = 0.025$) and with the strength of host association (i.e., the absolute host-association value; $\rho = 0.70$, $p = 0.021$); these two predictors were themselves strongly correlated ($\rho = 0.73$), so their effects cannot be disentangled. In contrast, predictive performance was not related to the signed host-association axis ($\rho = -0.25$, $p = 0.45$) and did not differ significantly between nematodes and protozoa (Wilcoxon $p = 0.32$; median cross-validated $R^2 = 0.43$ and 0.11 , respectively). Overall, rarer and more host-generalist parasites tended to be less predictable.

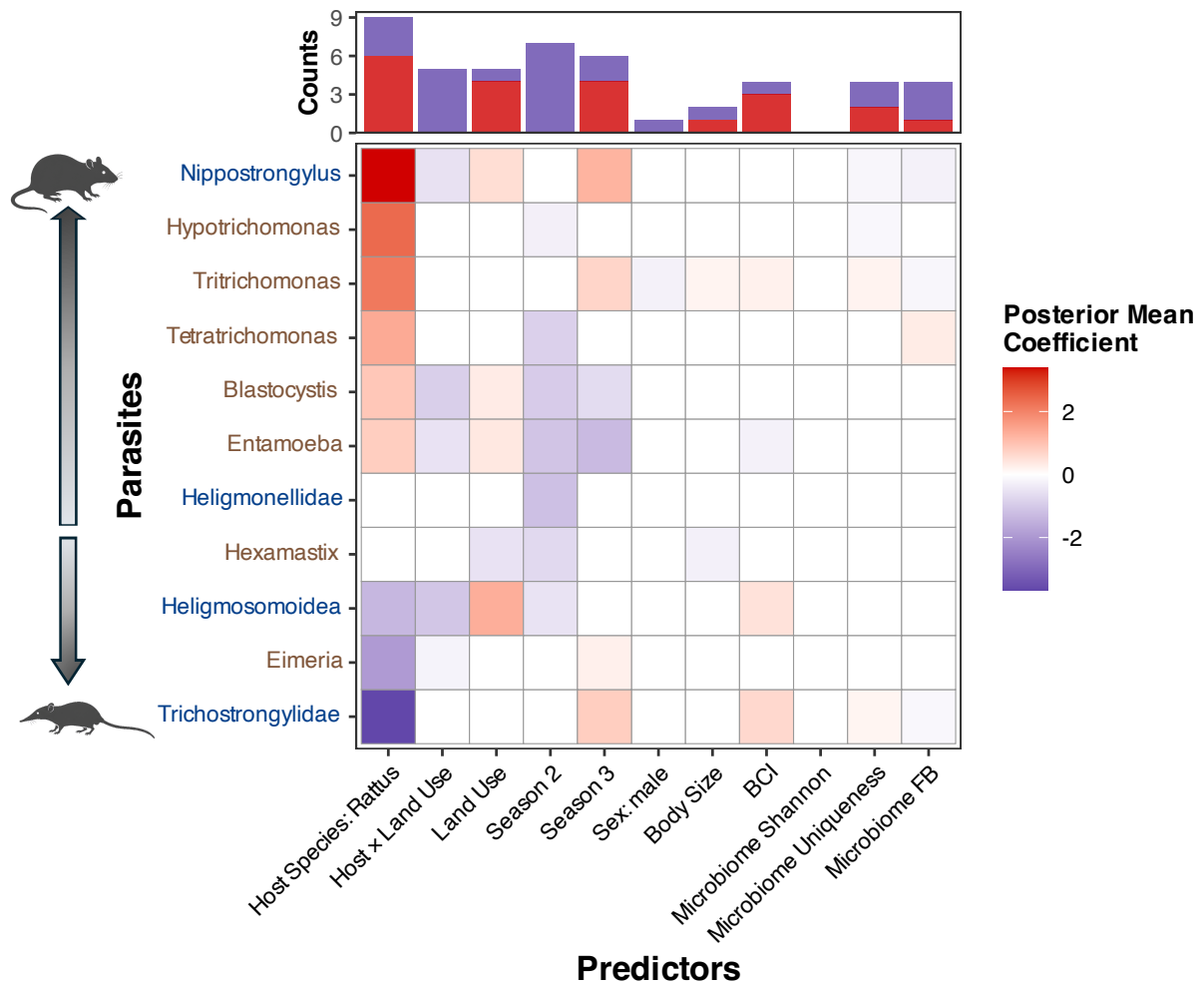


Figure S7: Responses of parasite taxa to host and environmental predictors. Bottom: Parasite-specific responses, where each row represents a unique parasite and columns represent predictors. Colors represent the mean posterior of the regression coefficients from the HMSC model: red indicates a significant positive association, purple indicates a significant negative association, and white indicates non-significant associations, defined as cases where the 90% posterior credible interval includes zero. Parasite names are color-coded by taxonomic group (blue: nematode; brown: protozoan). Categorical predictors are shown as deviations from their reference level: host species is the effect of *R. rattus* relative to the reference *M. brevicaudata*; Season 2 and Season 3 are relative to Season 1; Sex is the effect of males relative to females. **Top:** Frequency of significant associations across the parasite community, with the bar height indicating the total number of parasites significantly responding to each predictor. Red indicates a significant positive association, and purple indicates a significant negative association.

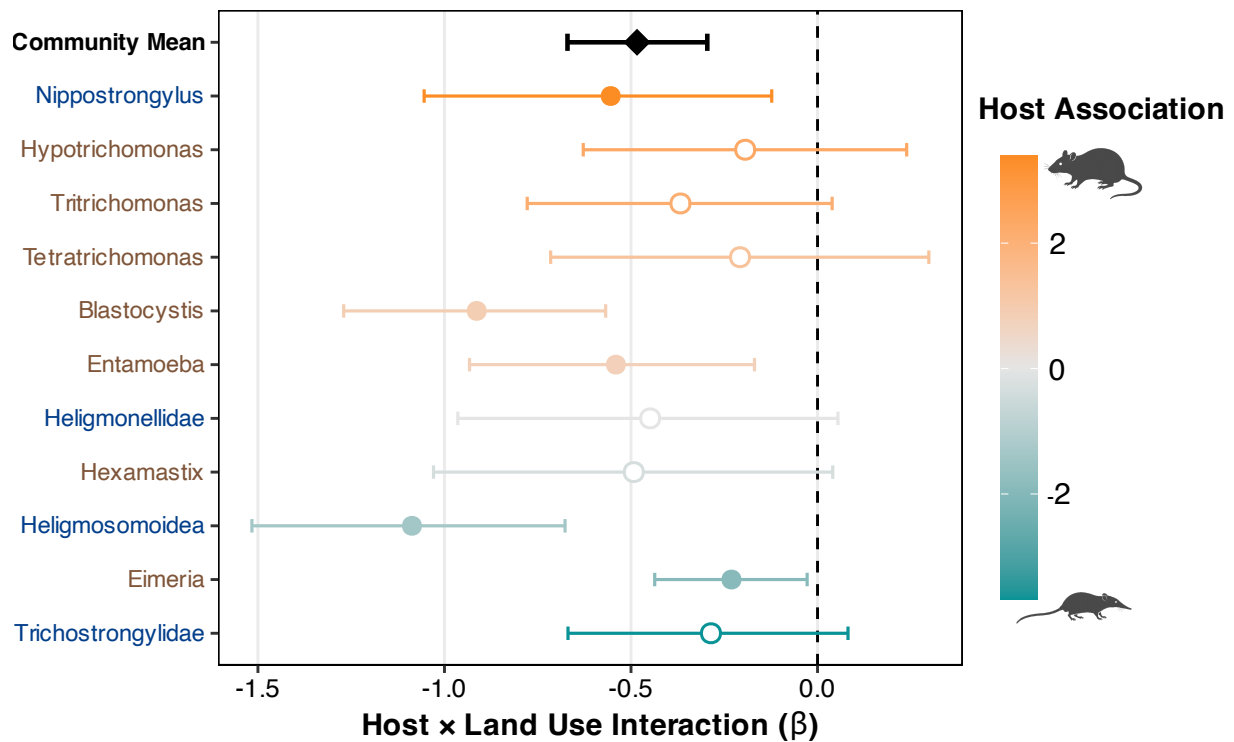


Figure S8: Per-parasite host × land-use interactions. Interaction coefficients (β) from the HMSC model, quantifying how each parasite's response to the land-use gradient differs between the two host species. Negative values indicate that occurrence declined more steeply (or increased less) toward forested habitat in *R. rattus* than in *M. brevicaudata*. Points are posterior means, and horizontal bars are 90% credible intervals; filled points denote interactions supported at the 90% level (intervals excluding zero), and open points those that were not. The black diamond (top) is the community-mean interaction across parasites. Points are colored by host association, the parasite's tendency to occur in one host over the other (teal, associated with *M. brevicaudata*; orange, associated with *R. rattus*). Parasite names are colored by taxonomic group (blue, nematode; brown, protozoan). Interaction coefficients were consistently negative across parasites, varying in magnitude rather than direction.

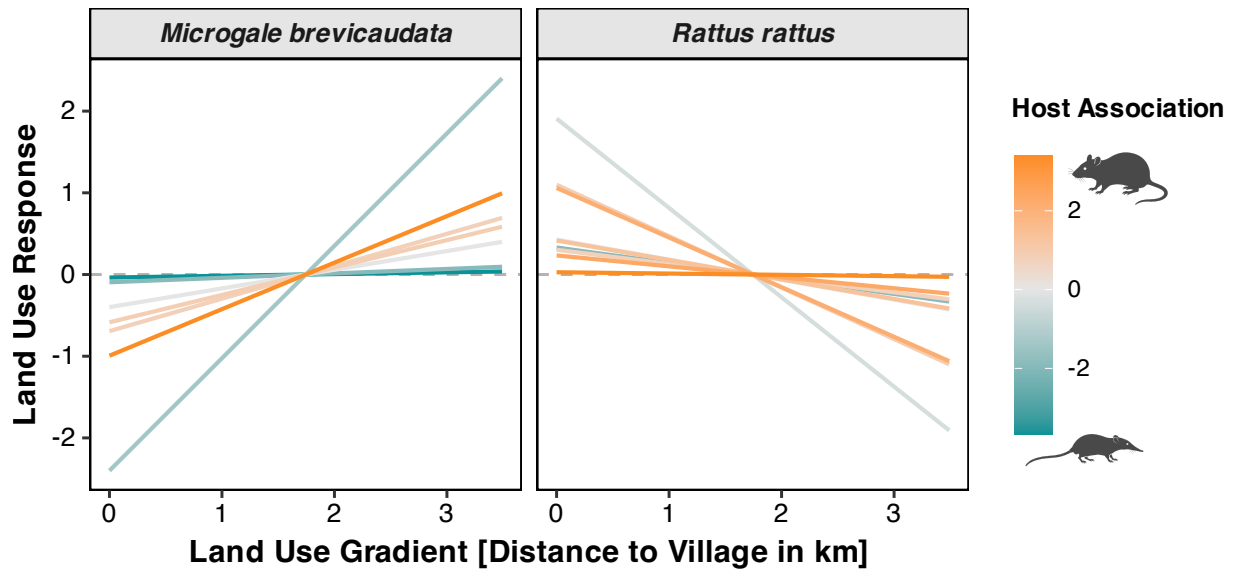


Figure S9: Latent-scale parasite responses to the land-use gradient in each host. Each line shows a parasite's response to the land-use gradient (distance to village in km; higher values indicate more forested habitat) on the model's latent (linear predictor) scale, shown separately for the native *M. brevicaudata* (left) and non-native *R. rattus* (right). Responses are centered so that each line passes through zero at the gradient midpoint. A line's slope is therefore its land-use response in that host (positive, occurrence increasing toward forested habitat; negative, decreasing), and its sign and steepness are directly comparable across parasites regardless of overall prevalence. Lines are colored by host association (teal, associated with *M. brevicaudata*; orange, associated with *R. rattus*). Only parasites present in at least 5 individuals of the corresponding host are shown. Because the latent scale is unbounded, these responses are not compressed by the ceiling and floor of the occurrence-probability scale (**Figure 4A**), confirming that the flatness of some high-prevalence taxa there reflects a genuinely weak response rather than saturation.

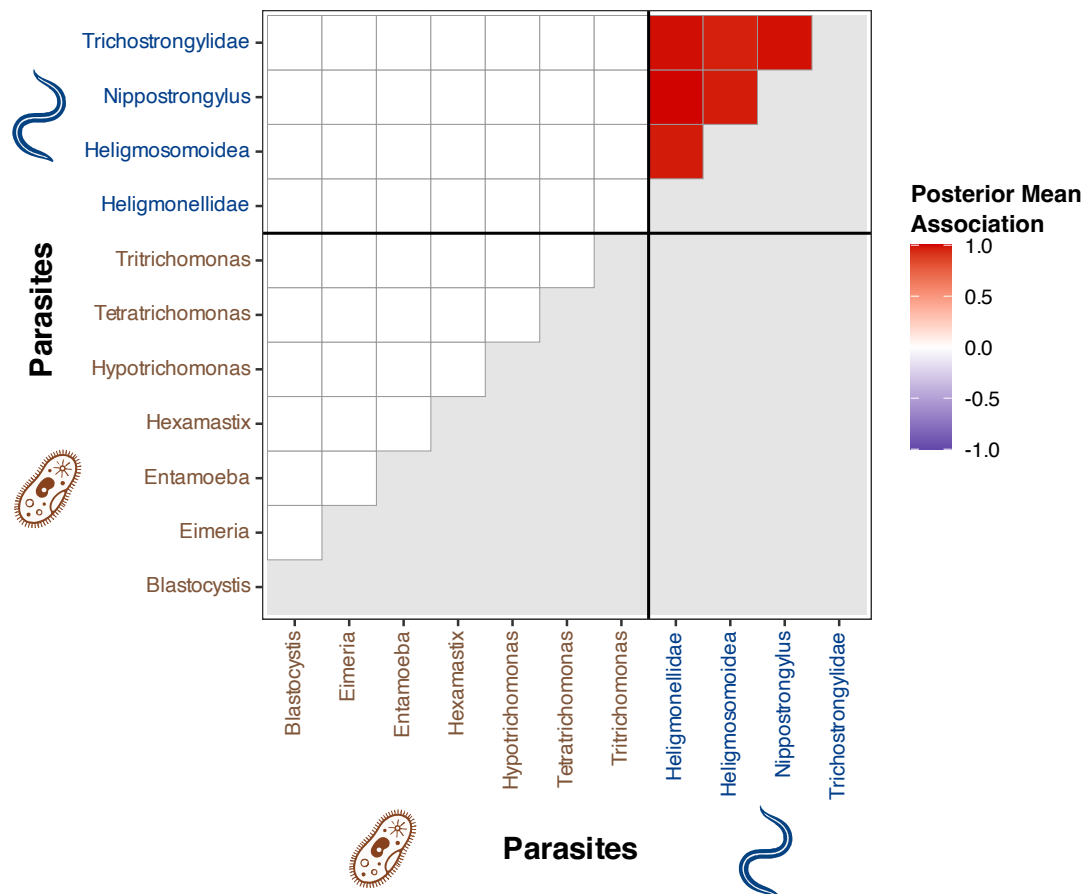


Figure S10: Parasite–parasite association network. Pairwise residual associations among parasite taxa estimated from the HMSC model (the Ω matrix at the individual random level), i.e., co-occurrence within hosts beyond what the measured predictors explain. Cell color shows the posterior mean association between a parasite pair: red indicates a well-supported positive association, purple a well-supported negative association, and white a pair whose association was not well supported (the 90% posterior credible interval includes zero). The lower triangle and the diagonal (grey) are masked, as each pair is shown only once in the upper triangle. Parasite names are colored by taxonomic group (blue: nematode; brown: protozoan) and ordered by taxonomic group. Well-supported positive associations were concentrated among nematode pairs.

SI note 1: Study site and small mammal sampling

Small mammals were sampled in the SAVA Region of northeastern Madagascar, in the vicinity of the village of Mandena (14.477049° S, 49.8147° E), with one site inside Marojejy National Park. Sampling at Mandena was conducted between October 2019 and September 2020. Seven sites were sampled along a degradation gradient: (1) semi-intact forest inside the national park, (2) secondary forest, (3) *savoka* (brushy regrowth after forest clearing), (4) agroforest (vanilla plantation), (5) mixed agriculture (sugarcane/coffee plantation), (6) rice fields, and (7) the village itself. Sites were located a minimum of ~500 m apart.

For sampling small mammals, a 100 m X 100 m grid of 121 live traps (11x11) was established, including 97 Sherman (H. B. Sherman Traps, Inc., Tallahassee, Florida, model LFA and XLK), and 24 Tomahawk (Tomahawk Live Trap, Hazelhurst, Wisconsin, model 201), placed 10 m apart and baited daily in the late afternoon with peanut butter. Additionally, two pitfall lines were installed between 20-50 m outside of the grid, running in parallel to the grid edge. Each pitfall line was 100 m in length, with 11 buckets dug into the ground and placed every 10 m, and an 80 cm high vertically oriented plastic fencing bisecting each bucket, stapled to vertical stakes, and a flange touching the ground covered with soil and leaf litter to block the passage of terrestrial small mammals and guide them towards a bucket. Each site was sampled for six consecutive nights across three sampling periods: October–November 2019 (before the wet season), March–April 2020 (end of the wet and hot season), and July–September 2020 (dry and cold season). Field climate data recorded by the trapping team (unpublished) indicate intermediate temperatures and the onset of rainfall in the first period, the highest temperatures and rainfall in the second, and the lowest values of both in the third.

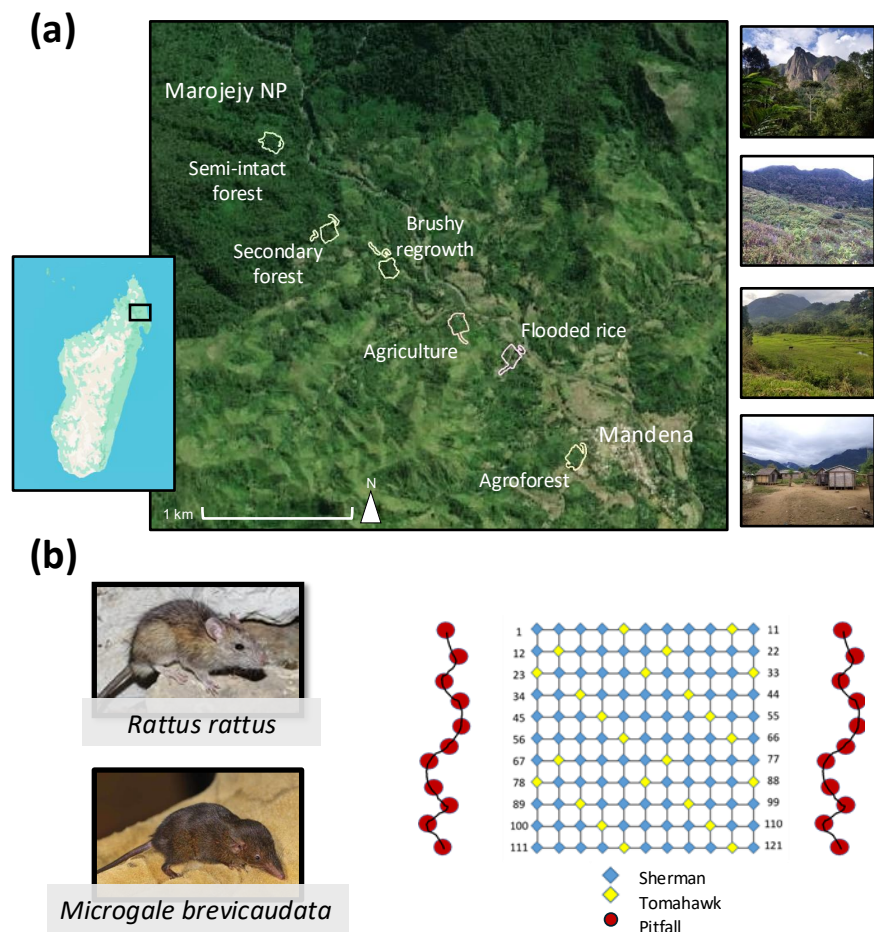


Figure S11: Study site and sampling scheme. (a) Sampling was conducted in northeastern Madagascar, in the village of Mandena near Marojejy National Park. Seven distinct land-use types were sampled. The images depict landscapes from top to bottom: semi-intact forest, *savoka*, rice fields, and village plots. (b) In each plot, an 11×11 trapping grid consisting of Sherman and Tomahawk traps was installed, along with two pitfall lines. A total of 333 individual *Rattus rattus* and 172 individual *Microgale brevicaudata* were captured.

SI note 2: Parasite detection, sequencing, and bioinformatic processing

We employed DNA metabarcoding to survey gut protozoan and nematode parasites. Fecal samples (~ 1 g) were collected and preserved in either nucleic acid preservation (NAP) buffer [1] or Zymo DNA/RNA Shield (Zymo Research, Irvine, California). The use of two storage solutions was necessary because of supply-chain constraints; mean sequencing read abundance did not differ significantly between sample types. Total DNA was extracted from all fecal samples using Zymo MiniPrep Fecal kits (Zymo Research, Irvine, California) following the manufacturer's protocol.

To screen for protozoan parasites, we targeted the 18S rRNA region using the G4 and G6 primer sets [2,3]. To screen for nematodes, we amplified the ITS2 rRNA region using the NC1/NC2 primer set [4]. Forward and reverse primers for each assay contained unique 8-nucleotide barcodes designed with a Hamming distance of at least 4. PCR reactions were performed in $15\ \mu\text{L}$ volumes containing $3\ \mu\text{L}$ of each forward and reverse primer ($2\ \mu\text{M}$ stock concentration), $7\ \mu\text{L}$ of mastermix, and up to $2\ \mu\text{L}$ template DNA (1–100 ng total). The mastermix consisted of $0.7\ \mu\text{L}$ Amplitaq Gold polymerase, $150\ \mu\text{L}$ MgCl_2 , $150\ \mu\text{L}$ Amplitaq Gold buffer, $12\ \mu\text{L}$ BSA, $6\ \mu\text{L}$ DMSO, and $344\ \mu\text{L}$ water. Cycling conditions included an initial 10-minute hot-start activation, followed by 35 cycles of denaturation for 15 s at 95°C , annealing for 30 s at 57°C for G4–G6 and 55°C for NC1/NC2, and extension for 40 s at 72°C , followed by a final 5-minute extension at 72°C .

Following PCR, DNA concentrations were measured, and amplicons were pooled, normalized, and purified using MinElute columns for multiplexed sequencing. The final library was sequenced three times on an Illumina MiSeq platform (v3 chemistry, 2×300 bp, 25 M reads) at the UC Davis Genome Center. Sequences were demultiplexed using *cutadapt* v3.4 with zero error tolerance [5].

Raw reads from both parasite assays were processed using the *dada2* pipeline [6]. Processing steps included filtering and trimming amplicons, removing PhiX reads, error correction, dereplication, inferring amplicon sequence variants (ASVs) using the pseudo-pooling method, paired-read merging, and chimera removal. Amplicons shorter than 100 bp were removed, and 15% PhiX reads were excluded. For both parasite groups, we calculated the relative read abundance of each ASV in each sample and removed ASVs that accounted for less than 1% of a sample's total relative read abundance to reduce the influence of sequencing artifacts and tag-jump errors. Samples with fewer than 500 total reads across both G4 and G6 assays ($n = 4$) were excluded from the protozoan dataset. For nematodes, samples below this threshold were retained but classified as uninfected.

Taxonomic assignment was performed separately for protozoa and nematodes. Nematode ASVs were identified using the *assignTaxonomy* function in *dada2*, with a minimum bootstrap threshold of 50, against the nemabiome ITS2 reference database v1.6.0. Protozoan ASVs were identified using a consensus approach. We first assigned taxonomy using both the *assignTaxonomy* function in *dada2* and the *IdTaxa* function in the *DECIPHER* package in R [7], using the SILVA non-redundant database v132 clustered at 99%. When the two approaches produced mismatched identifications, ASVs were queried against the NCBI GenBank database to assign a final consensus taxonomy.

Because the two parasite groups differed in taxonomic resolution, we applied group-specific procedures before downstream analysis. For protozoa, ASVs were first filtered taxonomically to retain only taxa established or suspected to be mammalian parasites, based on reference

literature and taxonomic databases, and the retained ASVs were then aggregated at the genus level. For nematodes, where taxonomic resolution was lower, we instead defined operational units by sequence clustering. ASVs were clustered into OTUs at 97% similarity using the *Clusterize* function in the *DECIPHER* package, and each OTU was assigned the taxonomy of its representative (centroid) ASV. To reduce over-splitting of closely related ITS2 variants into separate OTUs, we curated the OTUs with the LULU algorithm [8]. LULU compares each OTU against all more-abundant OTUs and merges it into a more-abundant "parent" OTU when three conditions are jointly met: the two share at least 84% pairwise sequence identity; the less-abundant OTU co-occurs with the parent in at least 90% of the samples in which it is detected; and their abundances are consistent with a parent-variant relationship (an average parent-to-daughter abundance ratio of at least 0.5 across co-occurring samples). Two curated OTUs that co-occurred in 503 of 505 sampled hosts were combined into a single unit for the downstream statistical analyses. Such near-complete co-occurrence makes the two OTUs effectively collinear, which would destabilize models in which the OTUs are entered as response variables.

Importantly, each nematode OTU was labeled with the lowest taxonomic rank to which its representative sequence could be confidently assigned, so unit names range from genus to superfamily. A family-level name denotes a single OTU that could not be resolved further, rather than the family as a whole. Names that appear nested, such as *Nippostrongylus* and Heligmonellidae, therefore refer to separate, non-overlapping OTUs. Protozoan units, by contrast, are genera, each aggregating all ASVs assigned to it. The two groups are thus delimited differently, and the breadth of a nematode OTU and of a protozoan genus are not equivalent.

Finally, to improve model robustness and predictive performance, we removed parasite taxa detected in fewer than 10 host individuals. The final dataset used for downstream analyses included 333 *R. rattus* and 172 *M. brevicaudata* individuals. Under the final classification scheme, these hosts were infected by seven protozoan and four nematode taxa (**Figure S2**).

SI note 3: Measuring environmental, host attributes, and host gut microbiome variables

To explore determinants of parasite infection patterns, we measured environmental conditions (land-use and seasonality), host attributes (body size, body condition, and sex), and gut microbiome structure (diversity, uniqueness, and the Firmicutes/Bacteroidota ratio) (**Table S1**). Host attributes, each individual's capture location, and gut microbiome variables are specific to the individual host, whereas season is shared among all hosts captured at a site during a given period.

Land-use gradient

We represented the land-use gradient with a single variable, the distance from each individual's capture location to the village center, and we validated this proxy against two descriptors of the environment: vegetation structure and small-mammal community composition.

Distance to village: We used the distance to the center of Mandena as a proxy for the land-use gradient, integrating gradients of habitat modification and human activity. For each captured individual, distance was measured with a GPS as the distance from the village center to the trap at which the animal was captured. We used this individual capture-location distance rather than a site-level average, so that the measure reflects each host's position along the disturbance gradient. To confirm that this single variable adequately captures the broader land-use gradient, we characterized vegetation structure and small-mammal community composition (see below) and tested their associations with distance.

Vegetation structure: We measured, with the assistance of a specialist botanist, habitat attributes in 16 plots (5 m × 5 m) within the sampling grid at each site, conducting measurements three times (over three seasons) during the sampling period. At each plot, we assessed eight habitat characteristics: (1) number of trees, (2) number of dead logs, (3) tree diameter at breast height, (4) tree height, (5) percent canopy cover, (6) number of lianas, (7) herbaceous vegetation height, and (8) percent herbaceous vegetation cover. We averaged measurements across all plots to obtain mean values for each site and season and summarized habitat variation among sites with a principal component analysis (PCA); prior to analysis all variables were centered at 0 and rescaled to unit variance. The first two principal components explained 86.66% of the variation across sites (PC1: 66.08%, PC2: 20.58%) (**Figure S12**). Distance to village was strongly correlated with vegetation PC1 (Spearman $\rho = 0.74$, $p < 0.001$) and with PC2 ($\rho = -0.50$, $p < 0.001$), confirming that increasing distance tracks the transition from disturbed to more natural habitat.

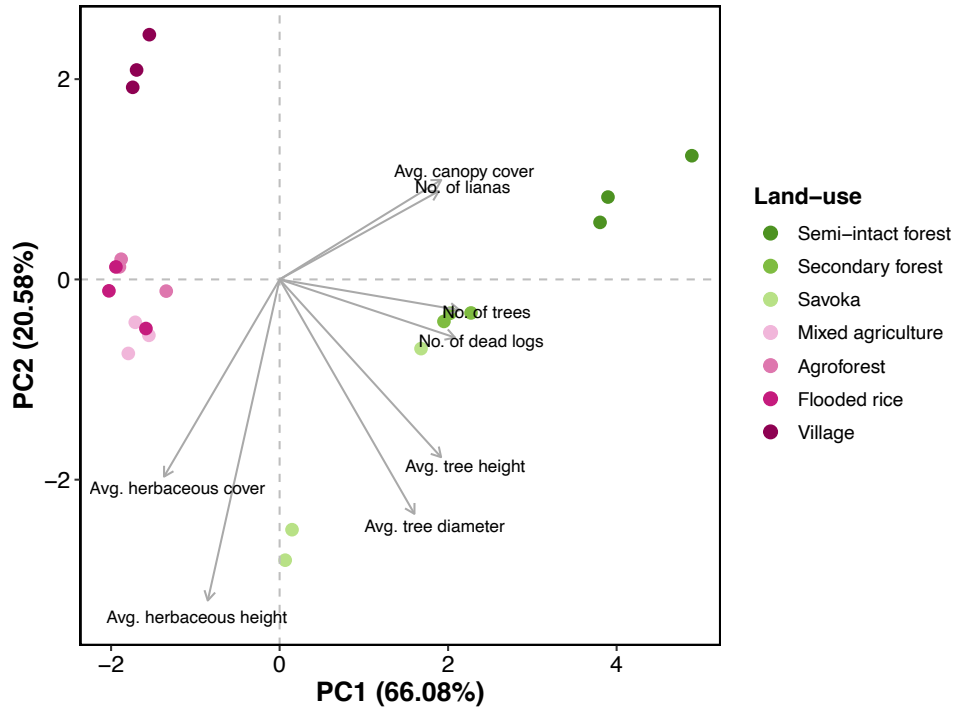


Figure S12: Vegetation PCA across sites. The first two principal components (PC1 and PC2) from the PCA of vegetation attributes. Each point represents a site from a single season, with color indicating the land-use type. Arrow length and direction indicate the contribution of each vegetation variable to the first two PCs.

Small-mammal community: Small-mammal community composition reflects habitat quality and disturbance and therefore provides an independent, biologically integrated descriptor of the land-use gradient. For each site and season, we recorded the capture abundance of each small-mammal species and summarized community composition with a PCA of the square-root-transformed abundances. The first two principal components explained 90.87% of the variation (PC1: 83.79%, PC2: 7.09%) (**Figure S13**). Distance to village was correlated with small-mammal community PC1 (Spearman $\rho = 0.35$, $p < 0.001$) and with PC2 ($\rho = 0.74$, $p < 0.001$), further supporting distance as a robust proxy for the land-use change gradient.

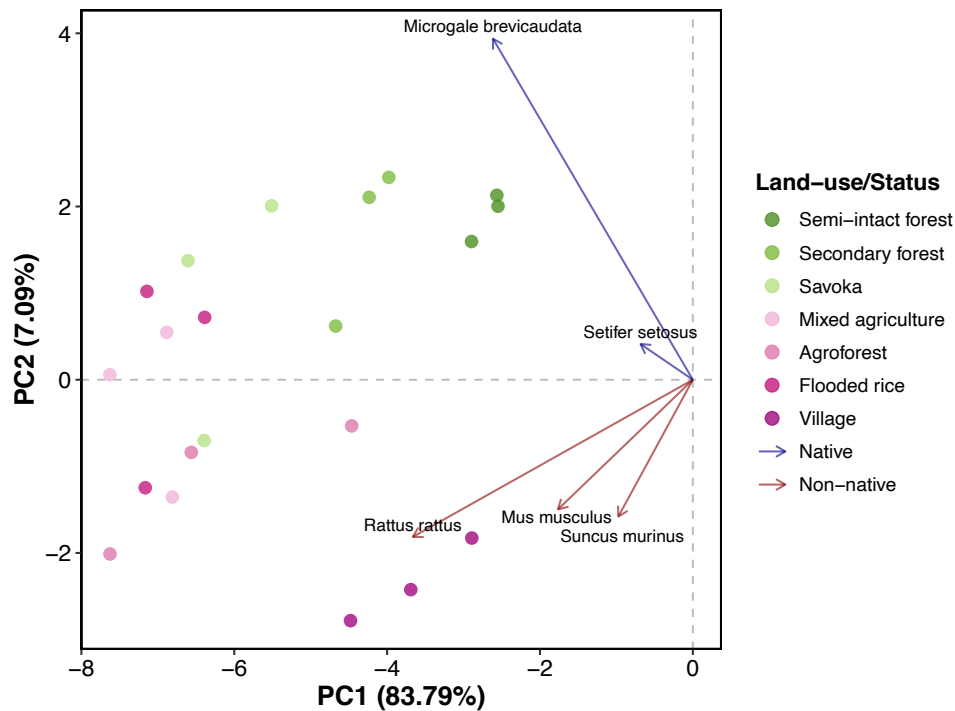


Figure S13: Small mammal community PCA across sites. The first two principal components (PC1 and PC2) from the PCA of small mammals species. Each point represents a site from a single season, with color indicating the land-use type. Arrow length and direction indicate the contribution of each species to the first two PCs. The arrow colors indicate native (blue) and non-native (red) species. For clarity, only the three species with the highest loadings on each PC axis are shown.

Host attributes

We characterized each individual host with three attributes: structural body size, body condition, and sex. Because *R. rattus* and *M. brevicaudata* differ markedly in body size, all continuous attributes were standardized (z -scored) within each species, so that they reflect intraspecific variation rather than differences between the two focal species.

Body size: We used hindfoot length as a measure of structural body size, log-transformed and standardized within each species. Hindfoot length is an external measurement that, unlike body mass or head-body length, is largely unaffected by short-term changes in condition, gut content, or measurement stretch, making it a repeatable proxy for an individual's structural size and, indirectly, its developmental stage. All external measurements were taken by a single collector, so between-observer variation does not contribute to this variable.

Body condition: We quantified body condition as the residual of a linear regression of log body mass on log hindfoot length, fitted separately within each species. This index captures body mass relative to structural size—an individual's energetic state—and, because it is taken as the regression residual, it is statistically independent of body size by construction (**Figure S14**).

Sex: We recorded sex (male or female) for each individual. Sex is a well-established correlate of host behavior, ranging, and immune investment, and was included to capture sex-linked differences in exposure and susceptibility to infection.

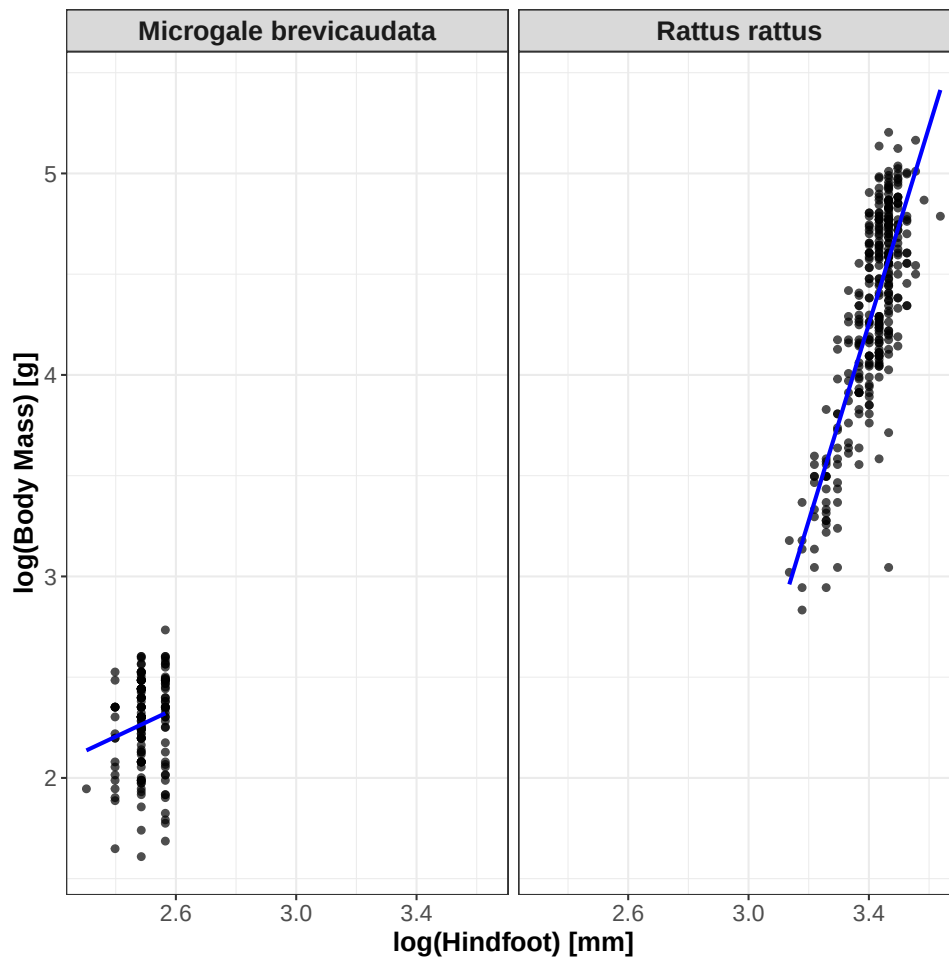


Figure S14: Body condition index (BCI). Relationship between log body mass and log hindfoot length for each focal host species (*Rattus rattus* and *Microgale brevicaudata*). Points are individual hosts, and the line is the within-species linear regression of log mass on log hindfoot length. BCI is defined as the residual from this regression—the vertical deviation of each individual from its species’ line—so that positive values indicate greater body mass than expected for a given structural size.

Gut bacterial microbiome

DNA was extracted from ~1g feces collected from trapped small mammals using Zymo Quick-DNA Fecal/Soil Microbe Miniprep kits (cat #D6010) using manufacturer protocols. 16S metabarcoding was conducted using 515F–806R primers to target the V4 region of the 16S SSU rRNA [9]. Each primer included an Illumina adapter, barcode, primer pad, and linker. Reactions were carried out in 25 μ L volumes consisting of 10 μ L of 1.25 μ M forward and reverse primer, 2 μ L of DNA, and 13 μ L of Platinum Hot Start PCR mastermix (ThermoFisher Scientific, cat #13000014). Reaction conditions were as follows: 95°C for 3min, 35x 98°C for 30secs, 58°C for 30secs, and 72°C for 30 s, followed by a final extension at 72°C for 5min. Concentrations were measured using Promega One Quantifluor kits on a Tecan plate reader. Samples were then normalized to 7 ng/ μ L prior to pooling. The product was cleaned using magnetic beads (bead:DNA ratio was 0.8:1) and sequenced at UC Santa Barbara Biological Nanostructures Laboratory on an Illumina MiSeq (v3 chemistry, 2x300 bp, 24M reads).

Sequences were demultiplexed using cutadapt (v.3.4) with zero error tolerance [5]. We then performed quality filtering steps using the *dada2* package in R [6]. Specifically, we filtered and trimmed amplicons (minimum length = 100, 15% PhiX removed), inferred and removed errors,

dereplicated sequences, inferred amplicon sequencing variants (ASVs) using the pseudo-pooling method, merged pairs, and removed chimeras. We assigned taxonomic identifications to ASVs using the *assignTaxonomy* function in *dada2*, using the SILVA nr99 SSU reference database (v.138.1).

Prior to analysis we filtered the ASV table in three steps. First, we removed all non-bacterial ASVs and those classified as ‘Chloroplast’ or ‘Mitochondria’. Second, we discarded ASVs that did not reach a relative abundance of 0.1% in at least two samples, thereby removing both within-sample rare variants and ASVs restricted to a single sample. Third, we excluded 26 samples with fewer than 5,000 total reads. These procedures retained 5,361 ASVs from the 19,400 detected across the two focal host species.

A phylogenetic tree of the retained ASVs was inferred from a multiple-sequence alignment (*DECIPHER*; [7]) using approximate maximum-likelihood under a GTR+ Γ model (FastTree; [10]) and midpoint-rooted.

To characterize the host gut microbiome as a proxy for physiological status and health, we summarized each individual’s community along three complementary dimensions: overall diversity, compositional individuality, and the balance of dominant phyla.

Diversity was quantified using the Shannon index. Higher diversity is generally associated with a stable, colonization-resistant gut community. Compositional individuality was quantified as microbiome uniqueness [11]: the mean weighted UniFrac distance [12] from each host to all of its conspecifics, computed separately within each focal species. Higher uniqueness reflects a greater deviation from the species-typical community, a signature of dysbiosis and reduced community stability. Finally, the Firmicutes/Bacteroidota (F/B) ratio, a compositional balance related to diet and energy harvest, was calculated after aggregating ASV reads at the Phylum level, using a pseudocount equal to the 0.1% detection threshold to avoid division by zero and a $\log_{10}$ transformation to normalize its distribution.

To account for inherent differences in baseline microbiomes between the two focal host species, all three microbiome metrics were standardized using Z-score scaling within each host species:

$$Z_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j}$$

where $x_{i,j}$ represents the metric value for individual i of host species j , and μ_j and σ_j are the species-specific mean and standard deviation, respectively. These standardized indices were subsequently used as predictors of the “Microbiome” component in our downstream variance-partitioning models.

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